7.5 Rotating in Three Dimensions

WORKSHEET

The question

Let $\mathbf{n}$ be a vector of unit length in three dimensions. For each vector $\mathbf{r}$, $\mathrm{f}(\mathbf{r})$ is defined by $\mathrm{f}(\mathbf{r}) = \mathbf{n} \times \mathbf{r}$.

(i) Given that

$$\mathbf{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \quad \text{and} \quad \mathbf{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix},$$

show that the x-component of $\mathrm{f}(\mathrm{f}(\mathbf{r}))$ is $-x(b^2 + c^2) + aby + acz$. Show further that

$$\mathrm{f}(\mathrm{f}(\mathbf{r})) = (\mathbf{n}.\mathbf{r})\mathbf{n} - \mathbf{r}.$$

Explain, by means of a diagram, how $\mathrm{f}(\mathrm{f}(\mathbf{r}))$ is related to $\mathbf{n}$ and $\mathbf{r}$.

(ii) Let R be the point with position vector $\mathbf{r}$ and P be the point with position vector $\mathrm{g}(\mathbf{r})$, where g is defined by

$$\mathrm{g}(\mathbf{s}) = \mathbf{s} + \sin\theta\,\mathrm{f}(\mathbf{s}) + (1 - \cos\theta)\,\mathrm{f}(\mathrm{f}(\mathbf{s})).$$

By considering $\mathrm{g}(\mathbf{n})$ and $\mathrm{g}(\mathbf{r})$ when $\mathbf{r}$ is perpendicular to $\mathbf{n}$, state, with justification, the geometric transformation which maps R onto P.

(iii) Let R be the point with position vector $\mathbf{r}$ and Q be the point with position vector $\mathrm{h}(\mathbf{r})$, where h is defined by

$$\mathrm{h}(\mathbf{s}) = -\mathbf{s} - 2\,\mathrm{f}(\mathrm{f}(\mathbf{s})).$$

State, with justification, the geometric transformation which maps R onto Q.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself What does $\mathbf{n} \times \mathbf{r}$ look like geometrically, when $\mathbf{r}$ is perpendicular to $\mathbf{n}$? And when $\mathbf{r}$ is parallel to $\mathbf{n}$?

STAGE 1 Crossing twice

Ask yourself Write $\mathbf{r} = \mathbf{r}_\parallel + \mathbf{r}_\perp$, where $\mathbf{r}_\parallel = (\mathbf{n}.\mathbf{r})\mathbf{n}$ is the component along $\mathbf{n}$. What is $\mathrm{f}(\mathrm{f}(\mathbf{r}))$ in terms of these?

STAGE 2 The rotation

Ask yourself Compute $\mathrm{g}(\mathbf{n})$. Then compute $\mathrm{g}(\mathbf{r})$ for $\mathbf{r} \perp \mathbf{n}$, using $\mathrm{f}(\mathrm{f}(\mathbf{r})) = -\mathbf{r}$ for such $\mathbf{r}$.

STAGE 3 The reflection

Pause and try Simplify $\mathrm{h}(\mathbf{s}) = -\mathbf{s} - 2\mathrm{f}(\mathrm{f}(\mathbf{s}))$ using part (i). What does it do to vectors along $\mathbf{n}$, and to vectors perpendicular to $\mathbf{n}$?

Try this next

1. **Rotations in the plane.** (From STEP III, 2017.) Show that rotation by θ about the point a in the complex plane sends z to $e^{\mathrm{i}\theta}z + a(1 - e^{\mathrm{i}\theta})$. Show that rotating by θ about a and then by ϕ about b is a rotation by $\theta + \phi$ about some point c, provided $\theta + \phi$ is not a multiple of 2π. What is it when $\theta + \phi = 2\pi$?

2. **A specific rotation.** Find the matrix of the rotation by 120° about the axis in the direction $(1, 1, 1)$. Show that it permutes the coordinate axes cyclically.

3. **Powers of $\mathbf{K}$.** Show that $\mathbf{K}^2\mathbf{r} = (\mathbf{n}.\mathbf{r})\mathbf{n} - \mathbf{r}$ and $\mathbf{K}^3 = -\mathbf{K}$, and deduce Rodrigues' formula from the exponential series.

4. **Two reflections make a rotation.** Show that reflecting in two planes through the origin, at angle α to each other, is a rotation by 2α about their line of intersection.

Further reading

- **Simon Altmann, *Rotations, Quaternions, and Double Groups*** (Dover). The definitive account of three-dimensional rotations, including Rodrigues' contribution.

- **Simon Altmann, "Hamilton, Rodrigues, and the quaternion scandal"**, *Mathematics Magazine* 62 (1989). A short, lively history.
- **John Stillwell, *Naive Lie Theory*** (Springer). Rotations, quaternions and Lie groups for undergraduates.

The full guided solution, and what the question leads to at university, is in Section 7.5:
https://keithmcnulty.github.io/step_book_mcnulty/ch07/rotations-3d.html